

Graphical Abstract

Director-CenterNet: Multi-Region Viewport Prediction for Automated Esports Broadcasting

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[**TODO**: Insert graphical abstract: pipeline diagram showing game-state tensor $\rightarrow$ Director-CenterNet $\rightarrow$ Main + PIP viewports overlaid on minimap.]

Highlights

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- Human observer attention in real-time strategy games is shown to be multimodal, not unimodal.
- Automatic observing is reformulated as multi-region viewport prediction for Esports broadcasting.
- Minority observer attention, previously discarded as noise, becomes supervision for auxiliary regions.
- Repulsion and smoothness objectives remove redundant overlap and camera thrashing.
- Predicted regions are presentation-agnostic and support PIP, split-screen, and director assistance.

Director-CenterNet: Multi-Region Viewport Prediction for Automated Esports Broadcasting

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Abstract

Human game observers, who control in-game cameras and provide viewers with an engaging experience, are a vital part of electronic sports (Esports). In professional broadcasts multiple observers cooperate to operate several camera feeds at once, because a single view cannot cover the events that occur concurrently across a large map. Consequently, various methods to create automatic observers have been explored. However, these methods predict only a single region per frame. This is problematic because the attention of human observers is not unimodal: observers watching the same replay agree on only about half of their viewport area, and their attention separates further as the game progresses. Previous learning-based observers treated this disagreement as noise and collapsed it into a single consensus region, which discards the minority modes and causes the predicted camera to oscillate between competing regions across consecutive frames. In this paper, we overcome these problems by reformulating automatic observing as *Multi-Region Viewport Prediction* (MRVP) and solving it with a heatmap-based detection method, CenterNet, in a real-time strategy game (e.g., StarCraft). Instead of collapsing the modes of human attention, the proposed model, *Director-CenterNet*, recovers them: we rank the modes of the aggregated observer distribution by the number of observers they represent, and assign

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the dominant mode to a primary region and the remaining modes to auxiliary regions. The predicted regions are presentation-agnostic, so they can be rendered as a main screen with picture-in-picture inserts, as a split-screen layout, or as a ranked candidate set for a human director. We design four training objectives for this task: a *Human Consensus Match* loss, a *Ranked Mode Coverage* loss, a *Spatial Repulsion* loss, and a *Trajectory Smoothness* loss. Consequently, we show that our automatic observer outperforms current rule-based methods, human observers, and single-region learning-based observers, and that it covers concurrent situations that a single-region observer cannot.

Keywords: Esports, Automatic observer, Multi-viewport prediction, CenterNet, Saliency-guided learning, Spatial resource allocation, StarCraft, Behavior cloning

1. Introduction

Human game observers in electronic sports (Esports) fulfill a dual role: they must simultaneously identify the most strategically significant region of the map *and* coordinate multiple camera feeds to ensure that concurrent events are not missed [?]. In professional Esports productions, a team of observers typically manages one main broadcast camera together with several auxiliary picture-in-picture (PIP) channels, each covering a distinct map region. Automating this workflow is challenging for three reasons: (1) the number of concurrent high-priority events is dynamic; (2) the spatial relationships among events evolve rapidly; and (3) the subjective notion of “broadcast saliency” cannot be reduced to a single, well-defined objective [?].

Prior work on automatic Esports observers has focused on the *single-viewport* setting, where the goal is to select one rectangular window of fixed size from the game map at each frame. Rule-based approaches [?] define priority scores over hand-crafted events and switch the camera accordingly. Learning-based methods [?] instead collect observational data from human spectators and train a detector-style model with Region of Common Interest (ROCI) loss to imitate consensus human camera behaviour. Subsequent work added an auxiliary objective derived from fixed-kernel spatial priors, which improved agreement with aggregated human attention without changing inference-time outputs [?].

Despite these advances, we argue that the single-region setting is not merely restrictive but *mismatched to the supervision it is trained on*. The evidence for this mismatch is already present in previous studies. Observers watching the same replay share only about 52% of their viewport area with each other, and questionnaires show that they attend to different aspects of the match, such as building upgrades, resource expansion, or fighting [?]. Moreover, this disagreement is not constant over a match: in the early game the viewports of observers mostly overlap, but as the game progresses and significant events begin to occur at several locations at the same time, the observers view different scenes and only a few of their viewports overlap. Accordingly, the reported accuracy of every method, including the human average, decreases monotonically with game progression [? ?]. We therefore regard the roughly 50% agreement ceiling not as measurement noise to be averaged away, but as evidence that the target distribution is *multimodal*.

This interpretation explains the limitations of previous work. ROCI [?] and fixed-kernel auxiliary priors [?] are both devices for rescuing a single prediction on a multimodal target: they sharpen or re-weight the distribution so that one region can be selected from it. Such devices necessarily discard the minority modes, and they become least effective in exactly the mid-to-late game phase in which the modes separate. In addition, when two regions receive nearly equal support, a model trained to select one of them oscillates between them across consecutive frames.

We address both limitations by reformulating automatic observing as **Multi-Region Viewport Prediction (MRVP)**: at each frame, the observer produces an ordered set of K non-redundant regions of fixed size. Unlike the single-region approaches mentioned previously, this formulation does not collapse the modes of human attention but covers them, and it allows camera thrashing to be suppressed explicitly rather than implicitly. We emphasise that the predicted regions are *presentation-agnostic*. The same output can be rendered as a main screen with picture-in-picture inserts, as a split-screen or multi-view grid, or as a ranked candidate set presented to a human director in the manner of [?]. The prediction problem is therefore independent of the production layout eventually chosen.

Furthermore, we propose to obtain the supervision for the auxiliary regions from the human observational data itself, rather than from a hand-designed prior. Because the aggregated observer distribution already contains the secondary modes that previous work suppressed, we smooth this distribution, extract its local maxima, and rank the resulting modes by the

number of observers they represent. The dominant mode supervises the primary region and the remaining modes supervise the auxiliary regions. In other words, the minority attention that ROCI treated as an obstacle becomes the training signal for the additional regions.

To solve MRVP we propose **Director-CenterNet**, built on CenterNet [?]. CenterNet’s dense heatmap representation is well-suited to MRVP because (i) it simultaneously scores all candidate viewport centres, enabling multi-peak extraction without iterative proposal generation; (ii) the dense heatmap preserves the multimodal structure of the target, whereas an instance mask requires a discrete assignment that discards it; and (iii) local-maximum extraction naturally supports a variable number of regions, allowing an auxiliary region to be suppressed when no secondary mode is detected.

Our contributions are as follows.

1. We show that human observer attention in a real-time strategy game is multimodal and that its modes separate as the match progresses, and we argue that single-region observers are therefore mismatched to their own supervision (Section 3).
2. We formalise Multi-Region Viewport Prediction (MRVP) as a presentation-agnostic task and propose evaluation metrics for multi-region coverage, per-observer service, and temporal stability (Section 3).
3. We propose Director-CenterNet, a CenterNet-based multi-peak observer that predicts an ordered set of regions in a single forward pass, together with a Ranked Mode Coverage loss that supervises the auxiliary regions using minority observer attention rather than a hand-designed prior (Section 4.3.2).
4. We demonstrate that the proposed method outperforms existing rule-based methods, human observers, and single-region learning-based observers, and that it covers concurrent situations a single-region observer cannot (Section 5).

2. Background and Related Work

2.1. Single-Viewport Automatic Observers

Automatic game observation has been studied primarily in real-time strategy (RTS) games, where the viewport selection problem is most challenging due to large map size and high event concurrency [?]. Rule-based observers such as AIIDE and SSCAIT [?] define priority scores over game events

and teleport the camera to the highest-priority event at each decision step. While interpretable, these systems require manual event taxonomies and are brittle under novel strategies.

Joo et al. [?] proposed the first event-free learning-based observer for StarCraft. They framed viewport selection as an instance segmentation problem, treating the human observer’s viewport as the object to detect. Using Mask R-CNN [?] and a Region of Common Interest (ROCI) loss that amplifies the overlapping area among multiple human observers, their method outperforms both rule-based baselines and behaviour cloning. Subsequent work added an auxiliary objective derived from fixed-kernel spatial priors, which improved agreement with aggregated human attention without changing inference-time outputs [?]. Both studies report that accuracy declines as the match progresses and that the viewports of observers become divided in the later phases — the observation this work builds on.

The Observer framework [?] provides an open-source, reproducible pipeline for data collection, preprocessing, and model training on StarCraft: Brood War replays, which we use in this work.

2.2. Object Detection Architectures

Our work builds on two complementary detection paradigms. **Center-Net** [?] frames detection as keypoint heatmap regression: the network predicts a Gaussian heatmap peaked at each object’s centre, with offset and size regression heads for bounding box recovery. Multi-peak extraction by local-maximum search provides a natural mechanism for detecting a variable number of objects per image. **DETR** [?] and RT-DETR [?] reframe detection as set prediction via Hungarian matching, eliminating non-maximum suppression. We choose CenterNet over transformer-based detectors for MRVP because the heatmap formulation preserves the multimodal structure of our target and admits a variable number of outputs without modifying a query set.

2.3. Saliency and Multi-Region Detection

The problem of distributing a limited number of viewports over a dynamic scene relates to multi-salient region detection [?], active perception [?], and sensor placement [?]. Classical saliency models incorporate hand-designed priors — including centre bias and background priors — to guide attentional selection [? ? ?]. Our work differs in that salient regions are defined

by aggregated human attention over a structured multi-channel game-state tensor rather than by low-level visual features in natural images.

2.4. Temporal Stability in Video-Based Observers

Camera thrashing — rapid oscillation between competing salient regions — is a known failure mode in learning-based observers. Trajectory smoothness losses penalising inter-frame displacement have been applied in video object detection [?] and automated cinematography [?] to reduce temporal inconsistency. We adapt this approach to the MRVP setting by penalising the L2 displacement of predicted viewport centres across consecutive frames.

3. Problem Formulation

3.1. Input Representation

Following [? ? ?], at each game frame t we construct a multi-channel state tensor $\mathbf{X}_t \in \mathbb{R}^{C \times H \times W}$ on a tile grid of size $H \times W$. Channels encode semantically distinct entity categories for each player (workers, ground units, air units, buildings), terrain elevation, combined vision, and resource locations. Each tile corresponds to 32×32 pixels in the game world.

To incorporate short-term temporal context, we stack frames at a fixed stride Δ :

$$\mathbf{X}_t^{\text{stk}} = \text{concat}[\mathbf{X}_t, \mathbf{X}_{t-\Delta}, \dots, \mathbf{X}_{t-W\Delta}] \in \mathbb{R}^{(W+1)C \times H \times W}. \quad (1)$$

Unless otherwise stated we use $\Delta = 8$ and $W = 3$, yielding a four-frame stack.

3.2. Human Supervision and Aggregation

For each frame t and human observer $u \in \{1, \dots, U\}$, we record a viewport $V_t^{(u)}$ specified by its top-left tile coordinate $\mathbf{p}_t^{(u)} \in \mathcal{C}$, where

$$\mathcal{C} = \{(i, j) \mid 0 \leq i \leq H - h, 0 \leq j \leq W - w\} \quad (2)$$

is the set of feasible top-left positions for a viewport of fixed tile size (h, w) . We aggregate human viewports into a hard union set

$$\mathcal{H}_t = \bigcup_{u=1}^U V_t^{(u)} \quad (3)$$

and a soft crowd coverage map

$$a_t(\mathbf{q}) = \frac{1}{U} \sum_{u=1}^U \mathbf{1}[\mathbf{q} \in V_t^{(u)}], \quad (4)$$

which measures the fraction of observers whose viewport covers tile $\mathbf{q}$.

3.3. Ranked Modes of the Observer Distribution

Rather than reducing a_t to a single consensus point, we recover its modes. We smooth the coverage map with a Gaussian filter, $\tilde{a}_t = G_{\sigma_m} * a_t$, and take its local maxima above a relative floor $\theta \cdot \max \tilde{a}_t$. Candidates are visited in descending order of $\tilde{a}_t$ and greedily suppressed within a minimum separation D , which prevents a single broad region of agreement from producing several modes. Each surviving mode $\mathbf{m}$ is assigned a *support*, the number of observers whose viewport covers it:

$$n_t(\mathbf{m}) = \sum_{u=1}^U \mathbf{1}[\mathbf{m} \in V_t^{(u)}]. \quad (5)$$

Sorting by support in descending order gives the ordered sequence

$$\mathcal{M}_t = (\mathbf{m}_t^{(1)}, \mathbf{m}_t^{(2)}, \dots), \quad n_t(\mathbf{m}_t^{(1)}) \geq n_t(\mathbf{m}_t^{(2)}) \geq \dots, \quad (6)$$

which we refer to as the Top-1, Top-2, ... regions of human attention at frame t . The Top-1 region corresponds to the consensus region targeted by previous single-region work; the remaining entries are the minority modes that previous work suppressed.

Two properties of $\mathcal{M}_t$ matter for this study. First, its cardinality grows as a match progresses, because observer attention separates in the mid-to-late game. Second, the reliability of the lower-ranked entries depends on U : with $U = 5$, a mode supported by two observers is not separable from sampling variation, whereas a larger observer pool makes minority modes measurable. This is the reason we extend the dataset of [?] with additional participants per replay, as described in Section 5.

3.4. Multi-Region Viewport Prediction (MRVP)

Definition. Given a game-state tensor $\mathbf{X}_t^{\text{stk}}$, predict a set of K viewport positions $\{\hat{\mathbf{p}}_t^{(k)}\}_{k=1}^K \subset \mathcal{C}$ such that:

1. $\hat{\mathbf{p}}_t^{(1)}$ (main viewport) maximally overlaps the human consensus region $\mathcal{H}_t$;
2. $\hat{\mathbf{p}}_t^{(k)}$ for $k \geq 2$ (PIP viewports) cover salient regions not covered by the main viewport;
3. Predicted viewports are spatially diverse (non-redundant);
4. Predicted viewport positions vary smoothly across consecutive frames.

This formulation generalises single-viewport prediction (the special case $K = 1$) while connecting MRVP to the multi-salient region detection literature.

3.5. Evaluation Metrics

Intersection Ratio (IR)..

$$\text{IR}_t = \frac{|V(\hat{\mathbf{p}}_t^{(1)}) \cap \mathcal{H}_t|}{|V(\hat{\mathbf{p}}_t^{(1)})|} \quad (7)$$

Intersection@ δ ..

$$\text{I@}\delta_t = \mathbf{1}[\text{IR}_t \geq \delta], \quad \delta \in \{0, 0.3, 0.5\}. \quad (8)$$

Best-of- K Intersection@ δ (BoK@ δ).. Whether *any* predicted region reaches overlap δ , which is the multi-region analogue of the first-versus-second-suggestion comparison reported in [?]:

$$\text{BoK@}\delta_t = \mathbf{1} \left[\max_{k \leq K} \frac{|V(\hat{\mathbf{p}}_t^{(k)}) \cap \mathcal{H}_t|}{|V(\hat{\mathbf{p}}_t^{(k)})|} \geq \delta \right]. \quad (9)$$

Observer Coverage (OC@ δ).. The fraction of *individual* observers served by at least one predicted region, which measures how many of the disagreeing observers the output satisfies:

$$\text{OC@}\delta_t = \frac{1}{U} \sum_{u=1}^U \mathbf{1} \left[\max_{k \leq K} \frac{|V(\hat{\mathbf{p}}_t^{(k)}) \cap V_t^{(u)}|}{|V_t^{(u)}|} \geq \delta \right]. \quad (10)$$

Note that IR and I@ δ are computed on the primary region only and are therefore directly comparable to previously reported single-region results, whereas BoK@ δ and OC@ δ characterise the multi-region setting. The gap between I@ δ and BoK@ δ indicates how much of the human attention distribution is reachable by additional regions; OC@ δ indicates how many of the disagreeing observers the output actually serves.

[**TODO**: Insert architecture figure: stacked game-state tensor $\rightarrow$ ResNet-FPN backbone $\rightarrow$ heatmap / offset / size heads $\rightarrow$ ranked multi-peak extraction $\rightarrow$ ordered region set, with the four training objectives annotated on the ranked human attention modes.]

Figure 1: Overview of Director-CenterNet. The stacked game-state tensor is processed by a ResNet-FPN backbone. The heatmap head produces a dense score map over all candidate region centres, and ranked multi-peak extraction yields an ordered set of regions in a single forward pass. Training is supervised by the ranked modes of the aggregated observer distribution (Section 3.3): the Top-1 mode supervises the primary region and the remaining modes supervise the auxiliary regions.

Viewport Displacement (VD).. Average L2 displacement of the main viewport centre across consecutive frames, measuring temporal stability:

$$VD = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\bar{\mathbf{p}}_t^{(1)} - \bar{\mathbf{p}}_{t+1}^{(1)}\|_2, \quad (11)$$

where $\bar{\mathbf{p}}_t^{(k)}$ denotes the centre of the k -th predicted viewport at frame t .

4. Director-CenterNet

4.1. Architecture Overview

Director-CenterNet is a heatmap-based multi-peak detector that extends CenterNet [?] to the MRVP problem. The architecture comprises three components: (1) a ResNet-FPN backbone encoder that transforms the stacked game-state tensor into a spatial feature map; (2) CenterNet heads that regress a dense Gaussian heatmap of viewport centres together with offset and size maps; and (3) a ranked multi-peak extraction step. Fig. 1 provides an overview.

4.2. CenterNet Backbone and Heads

Input.. The stacked tensor $\mathbf{X}_t^{\text{stk}} \in \mathbb{R}^{(W+1)C \times H \times W}$ is fed to a ResNet-50 [?] backbone with a Feature Pyramid Network (FPN) [?] to obtain a feature map $\mathbf{F}_t \in \mathbb{R}^{C_f \times H_s \times W_s}$ at stride s . The tile map is consumed at its native resolution without resizing, and we take the finest FPN level, $s = 4$. For the 128×128 tile maps used here this yields a 32×32 heatmap, i.e. 1024 candidate region centres, with each heatmap cell covering a 4×4 tile block. Since a viewport spans 12×20 tiles, it occupies 3×5 cells; the stride is

therefore already coarse relative to the target, and we return to this point in Section 6.

Heatmap head.. Two 3×3 convolutions followed by a 1×1 sigmoid layer produce the region-centre heatmap $\hat{\mathcal{Y}}_t \in [0, 1]^{H_s \times W_s}$. The ground-truth heatmap is rendered by placing a 2D Gaussian at each ranked mode of Section 3.3:

$$\mathcal{Y}_t(\mathbf{q}) = \max_k \omega_k \exp\left(-\frac{\|\mathbf{q} - \pi(\mathbf{m}_t^{(k)})\|^2}{2\sigma^2}\right), \quad (12)$$

where $\omega_k = n_t(\mathbf{m}_t^{(k)})/U \in (0, 1]$ weights a mode by its normalised support. Using the support as an amplitude, rather than treating all modes identically, preserves the ranking in the target and prevents a mode observed by a single participant from being trained as strongly as the consensus region.

The operator $\pi(\cdot)$ maps a mode to the centre of its assigned heatmap cell, and it is not cosmetic. A mode at a continuous tile coordinate can fall up to $s\sqrt{2}/2$ tiles from the nearest sampled grid point, which for $\sigma = 2$ tiles and $s = 4$ attenuates the rendered peak by up to $\exp(-1) \approx 0.37$. Rendering at the continuous coordinate would therefore make the target amplitude encode quantisation error as much as observer support, defeating the ranking it is meant to carry; a mode with support 1 of $U = 5$ would render anywhere in $[0.074, 0.2]$ instead of exactly 0.2. We render at the assigned cell, as [?] does, and leave the sub-cell residual to the offset head.

Offset and size heads.. Following [?], a sub-pixel offset map $\hat{\mathbf{O}}_t \in \mathbb{R}^{2 \times H_s \times W_s}$ and a size map $\hat{\mathbf{Z}}_t \in \mathbb{R}^{2 \times H_s \times W_s}$ recover accurate region boxes from detected centres. Output convolutions of all three heads are initialised at a small standard deviation, following [?]; initialising them by the fan-out rule used for the hidden layers gives a standard deviation of $\sqrt{2}$ for a 1×1 convolution with a single output channel, which saturates the sigmoid and drives the offset head far outside its $[0, 1)$ target range, so that decoded centres fall off the map and are clamped to its border.

Ranked multi-peak extraction.. At inference a 3×3 max-pooling suppresses non-maxima, and the top- K peaks with score above a confidence threshold τ are returned in descending score order. The highest-scoring peak is the primary region. If fewer than K peaks exceed τ , fewer regions are returned, so the number of active auxiliary regions adapts to the number of concurrent events.

Two details of this step are load-bearing. First, τ must lie below the smallest meaningful target amplitude $1/U$, since by Eq. 12 a mode supported by one observer is trained toward exactly $1/U$; a threshold at or above that value makes minority regions unreachable no matter how well the model fits them. Second, `max_pool` pads with $-\infty$, so a cell on the heatmap border competes against fewer neighbours than an interior cell and becomes a local maximum with probability $1/4$ rather than $1/9$ under a flat heatmap. We therefore exclude the outermost ring of cells from the candidate set, which costs nothing reachable: a viewport at the map corner has its centre at $(h/2, w/2)$, one to two cells inside the boundary.

Differentiable region centres. The peak indices returned by a top- K selection are discrete and carry no gradient, so a centre formed as (peak cell + offset head) is a function of the regression heads alone and is independent of the heatmap. The repulsion and smoothness objectives below are statements about *where the model looks*, and defined on such a centre they cannot reach the map that decides it. We therefore define a soft centre by a softmax over the heatmap logits in a window of radius r around each peak,

$$\bar{\mathbf{p}}_t^{(k)} = s\left(\mathbf{c}_t^{(k)} + \sum_{\delta \in \mathcal{W}_r} \text{softmax}(z_t(\mathbf{c}_t^{(k)} + \delta)) \delta\right), \quad (13)$$

where $\mathbf{c}_t^{(k)}$ is the k -th peak cell, z_t the pre-sigmoid logits, and $\mathcal{W}_r$ the offsets within the window. This quantity is differentiable with respect to $\hat{\mathcal{Y}}_t$ and collapses toward the peak cell as the peak sharpens. It is used only by $\mathcal{L}_{\text{rep}}$ and $\mathcal{L}_{\text{smooth}}$; the regions emitted at inference remain the decoded (cell + offset) ones, since adding both displacements would double-count a sub-cell position that only $\mathcal{L}_{\text{off}}$ supervises.

4.3. Multi-Objective Training Losses

Director-CenterNet is trained with four objectives designed for this task, alongside the standard CenterNet regression losses. The four divide the heatmap between them: $\mathcal{L}_{\text{hcm}}$ owns the primary cell and the background, $\mathcal{L}_{\text{rmc}}$ owns the auxiliary modes, and $\mathcal{L}_{\text{rep}}$ and $\mathcal{L}_{\text{smooth}}$ act on the geometry of the decoded regions.

4.3.1. Human Consensus Match Loss $\mathcal{L}_{\text{hcm}}$

The primary region must be anchored to the Top-1 mode. Writing $\mathcal{Y}_t^{(1)}$ for the heatmap of Eq. 12 restricted to $\mathbf{m}_t^{(1)}$, and Ψ_t for the auxiliary support

set defined in Section 4.3.2, we apply the modified focal loss [?]:

$$\mathcal{L}_{\text{hcm}} = -\frac{1}{N} \sum_{\mathbf{q} \notin \Psi_t} \begin{cases} (1 - \hat{\mathcal{Y}}_t(\mathbf{q}))^{\alpha_f} \log \hat{\mathcal{Y}}_t(\mathbf{q}) & \text{if } \mathbf{q} = \pi(\mathbf{m}_t^{(1)}), \\ (1 - \mathcal{Y}_t(\mathbf{q}))^{\beta_f} \hat{\mathcal{Y}}_t(\mathbf{q})^{\alpha_f} \log(1 - \hat{\mathcal{Y}}_t(\mathbf{q})) & \text{otherwise,} \end{cases} \quad (14)$$

where N is the number of positive locations and α_f, β_f are focal loss hyperparameters.

Two aspects of Eq. 14 differ from a direct transcription of CornerNet and both concern the auxiliary modes. First, the penalty-reduction factor uses the *joint* target $\mathcal{Y}_t$ rather than $\mathcal{Y}_t^{(1)}$. In CornerNet the $(1 - \mathcal{Y})^{\beta_f}$ factor protects every ground-truth location from being pulled down as a hard negative; weighting by $\mathcal{Y}_t^{(1)}$ alone leaves the auxiliary modes unprotected, since $\mathcal{Y}_t^{(1)} \approx 0$ there and the factor is therefore ≈ 1 . The focal term would then be trained to erase precisely what $\mathcal{L}_{\text{rmc}}$ is trying to create. Second, down-weighting by itself is insufficient: at an auxiliary mode with support 1 of $U = 5$ the joint target is 0.2, so $(1 - 0.2)^4 = 0.41$ still leaves substantial downward pressure. We therefore exclude Ψ_t from this loss entirely and delegate it to $\mathcal{L}_{\text{rmc}}$, so that the two objectives are complementary by construction rather than by assumption.

4.3.2. Ranked Mode Coverage Loss $\mathcal{L}_{\text{rmc}}$

The auxiliary regions must cover the remaining modes. Let $\mathcal{M}_t^- = (\mathbf{m}_t^{(2)}, \dots, \mathbf{m}_t^{(K)})$ and let $\mathcal{Y}_t^-$ be the support-weighted heatmap of Eq. 12 rendered from $\mathcal{M}_t^-$ alone. Writing $\mathcal{A}_t^{(1)}$ for the binary mask of tiles already covered by the primary region, the auxiliary region outside it is

$$\Omega_t = \{\mathbf{q} : \mathcal{A}_t^{(1)}(\mathbf{q}) = 0\}, \quad (15)$$

and the subset of it that actually carries auxiliary mass is

$$\Psi_t = \{\mathbf{q} \in \Omega_t : \mathcal{Y}_t^-(\mathbf{q}) > \epsilon\}. \quad (16)$$

The loss is a masked mean-squared error over Ψ_t :

$$\mathcal{L}_{\text{rmc}} = \frac{1}{|\Psi_t|} \sum_{\mathbf{q} \in \Psi_t} (\hat{\mathcal{Y}}_t(\mathbf{q}) - \mathcal{Y}_t^-(\mathbf{q}))^2. \quad (17)$$

When $\mathcal{M}_t^-$ is empty, which occurs in early-game frames where observers agree, $\mathcal{L}_{\text{rmc}} = 0$ and no auxiliary supervision is applied.

Normalising over Ψ_t rather than over Ω_t is essential. Ω_t excludes only the primary region and therefore covers almost the whole heatmap — 1009 of 1024 cells in our configuration — while the auxiliary mass occupies roughly a dozen of them. Averaging over Ω_t dilutes the only informative cells by two orders of magnitude, whereas $\mathcal{L}_{\text{hcm}}$ normalises by the number of positives. Suppressing the true background is already $\mathcal{L}_{\text{hcm}}$ ’s role; this term only has to raise the prediction at the ranked minority modes to their support-weighted amplitude.

4.3.3. Spatial Repulsion Loss $\mathcal{L}_{\text{rep}}$

To keep the predicted regions non-redundant we penalise pairwise Intersection over Union between them, evaluated on the soft centres of Eq. 13:

$$\mathcal{L}_{\text{rep}} = \sum_{k < l} \text{IoU}(V(\bar{\mathbf{p}}_t^{(k)}), V(\bar{\mathbf{p}}_t^{(l)})). \quad (18)$$

Regions are shifted rather than cropped at the map boundary, matching the feasible set $\mathcal{C}$ of Eq. 2, so that this term always sees the fixed-size viewport the task is defined over and cannot reduce an overlap penalty by sliding a region partly off the map.

When fewer than two regions are returned, $\mathcal{L}_{\text{rep}} = 0$. Note that this term cannot create regions; it only separates regions that $\mathcal{L}_{\text{rmc}}$ has already produced. Its effect is therefore defined only conditional on $\mathcal{L}_{\text{rmc}}$, which is why the corresponding cell of the ablation lattice in Section 5.6 is left empty rather than measured.

4.3.4. Trajectory Smoothness Loss $\mathcal{L}_{\text{smooth}}$

To suppress camera thrashing we penalise inter-frame displacement of the primary region centre. Writing $d_t = \|\bar{\mathbf{p}}_t^{(1)} - \bar{\mathbf{p}}_{t+1}^{(1)}\|_2$ in tiles and $D = \sqrt{H^2 + W^2}$ for the map diagonal,

$$\mathcal{L}_{\text{smooth}} = \frac{1}{D(T-1)} \sum_{t=1}^{T-1} \mathcal{H}_{\Delta_s}(d_t), \quad \mathcal{H}_{\Delta_s}(d) = \begin{cases} \frac{1}{2}d^2 & d \leq \Delta_s, \\ \Delta_s(d - \frac{1}{2}\Delta_s) & d > \Delta_s. \end{cases} \quad (19)$$

The objective is applied to the primary region only, since the auxiliary regions are expected to be reallocated as events appear and disappear.

A squared- L_2 penalty in tile units, which is the direct transcription of the intent, makes “hold still” cheaper than “follow the action”. On a 128×128

Table 1: Summary of Director-CenterNet training losses.

Loss	Symbol	Type	Purpose
Human Consensus Match	$\mathcal{L}_{\text{hcm}}$	Focal	Anchor main viewport to human consensus
Offset regression	$\mathcal{L}_{\text{off}}$	L1	Sub-pixel centre accuracy
Size regression	$\mathcal{L}_{\text{size}}$	L1	Viewport size accuracy
Ranked Mode Coverage	$\mathcal{L}_{\text{rmc}}$	Masked MSE	Direct auxiliary regions to the remaining ra
Spatial Repulsion	$\mathcal{L}_{\text{rep}}$	IoU penalty	Keep the predicted regions non-redundant
Trajectory Smoothness	$\mathcal{L}_{\text{smooth}}$	Huber	Suppress thrashing of the primary region

map two independent predictions differ by $\mathbb{E}[d^2] \approx 5461$ at initialisation, several orders of magnitude above a focal term of $O(1)$; because the gradient norm is clipped globally, such a term does not merely outweigh the others but rescales the whole update and starves them. The Huber form caps the per-sample gradient at Δ_s so that small motion remains almost free while a large jump is expensive but bounded, and the division by D places the result on an $O(1)$ scale. The order matters: normalising before applying the Huber would put Δ_s at Δ_s/D and shrink the objective by roughly four orders of magnitude, disabling it rather than correcting it. We additionally hold λ_{sm} at zero for the first E_0 epochs and ramp it linearly to its full value by epoch E_1 , because a trajectory objective imposed on a randomly initialised heatmap is most cheaply satisfied by not moving at all.

4.3.5. Joint Training Objective

The total loss combines the standard CenterNet detection objective with the four multi-viewport losses:

$$\mathcal{L} = \mathcal{L}_{\text{hcm}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{sz}}\mathcal{L}_{\text{size}} + \lambda_{\text{rmc}}\mathcal{L}_{\text{rmc}} + \lambda_{\text{rep}}\mathcal{L}_{\text{rep}} + \lambda_{\text{sm}}\mathcal{L}_{\text{smooth}}, \quad (20)$$

where $\mathcal{L}_{\text{off}}$ and $\mathcal{L}_{\text{size}}$ are the standard CenterNet offset and size regression losses [?].

Table 1 summarises the role of each loss component.

5. Experiments

5.1. Dataset and Preprocessing

We use the same StarCraft: Brood War dataset and Observer preprocessing pipeline as [? ?]. Replay files were collected from online communities

Table 2: Structure of the aggregated observer distribution by game progression: mean number of extracted modes and mean support of the Top-1 to Top-3 modes. **[TODO: Fill in after data collection.]**

Progression	$ \mathcal{M}_t $	Top-1	Top-2	Top-3
1/4	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
2/4	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
3/4	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
4/4	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]

(Ygosu, BWReplays) with constraints on player APM (> 250), race matchups (Z vs. P, T vs. P, Z vs. T), competitive maps (Fighting Spirit, Jade, Othello), and starting locations. Human observational data were collected from participants (aged 20–35) who freely spectated replay files while viewport coordinates were recorded at each frame. A total of approximately 420,000 frames from 21 games were collected.

Extending the observer pool. The original collection recorded $U = 5$ observers *per replay* [?]. As argued in Section 3.3, five observers are sufficient to estimate the consensus region but not to distinguish a genuine minority mode from sampling variation: a mode supported by two of five falls within the uncertainty of the estimate, and by Eq. 12 the smallest representable amplitude is $1/U = 0.2$. We therefore recruit additional participants under the same protocol as [?], raising the pool to **[TODO: state final U ; target 10–15]** observers per replay. With a larger pool a mode supported by a third of the observers becomes measurable, which is the minimum requirement for supervising the auxiliary regions. Participant recruitment criteria, instructions, and post-session questionnaires follow [?] so that the extended data remain compatible with the original portion.

Characterising the target distribution. Before training we report the structure of the supervision itself: the number of extracted modes per frame as a function of game progression, the support distribution of the Top-1 to Top- N modes, and pairwise observer agreement as a function of U . Table 2 presents these statistics. This analysis is a contribution in its own right, since it quantifies the multimodality that motivates the task.

Game-state tensors follow the channelisation protocol of [?]: four entity channels per player (workers, ground units, air units, buildings), terrain

elevation, combined vision, and resource channels. Temporal stacking uses $\Delta = 8$, $W = 3$ (four-frame stack), and the 128×128 tile map is consumed at its native resolution.

5.2. Experimental Setup

Splits.. Three replay-level folds are used for cross-validation following [? ?]. All main comparison results are averaged over three folds and three random seeds $\{123, 456, 789\}$.

Implementation.. Backbone: ResNet-50 pretrained on ImageNet with FPN, finest level ($s = 4$), giving a 32×32 heatmap. Gaussian rendering: $\sigma = 2$ tiles, which matches the CenterNet radius heuristic for a 3×5 -cell target. Mode extraction: Gaussian smoothing $\sigma_m = 4$ tiles, relative floor $\theta = 0.35$, minimum separation $D = 12$ tiles. Loss weights: $\lambda_{\text{off}} = 1$, $\lambda_{\text{sz}} = 0.1$, $\lambda_{\text{rnc}} = 0.5$, $\lambda_{\text{rep}} = 0.3$, $\lambda_{\text{sm}} = 0.02$. Smoothness shaping: Huber knee $\Delta_s = 5$ tiles, warmup $E_0 = 5$, $E_1 = 10$. Soft-centre window radius $r = 2$ cells. Region set: $K = 3$, confidence threshold $\tau = 0.1$. Note that τ is set below $1/U$ for the reasons given in Section 4; the inference-time score filter is held at the same value, since a filter above τ would silently remove every minority region.

Optimiser.. SGD with momentum 0.9, weight decay 5×10^{-4} , learning rate 5×10^{-3} with StepLR decay (step 3, factor 0.1), batch size 16, 30 epochs.

5.3. Baselines

We compare Director-CenterNet against the following methods.

- **BC**: Behaviour cloning via MSE regression to human viewport coordinates.
- **AIIDE**: Rule-based observer [?].
- **SSCAIT**: Rule-based observer with event priorities and timers [?].
- **Mask R-CNN (1H)**: Single-viewport detection, one human label [?].
- **Mask R-CNN (5H)**: Single-viewport detection, five human labels [?].
- **Mask R-CNN + ROCI**: With Region of Common Interest loss [?].

Table 3: Main viewport performance on the standard test set (fixed race, map, starting location). Results averaged over three folds $\times$ three seeds. Best result in **bold**, second best underlined. **[TODO: Fill in experimental values.]**

Method	@any $\uparrow$	@0.3 $\uparrow$	@0.5 $\uparrow$	IR $\uparrow$
BC	0.061	0.059	0.046	–
AIIDE	0.524	–	–	–
SSCAIT	0.491	–	–	–
Mask R-CNN (1H)	0.593	–	–	–
Mask R-CNN (5H)	0.569	–	–	–
Mask R-CNN + ROCI	0.588	–	–	–
Mask R-CNN + prior	<u>0.824</u>	<u>0.798</u>	<u>0.770</u>	<u>0.709</u>
CenterNet (single)	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
Director-CenterNet (Ours)	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]

- **Mask R-CNN + fixed-kernel prior**: the auxiliary-prior variant of [?], the strongest previously reported single-region result and our primary point of comparison.
- **CenterNet (single)**: the proposed backbone trained with $\mathcal{L}_{\text{hcm}}$ only, i.e. the first row of Table 6. This isolates the effect of the backbone change from that of the multi-region formulation, and is therefore the same architecture as Director-CenterNet rather than a separate plain-backbone CenterNet, which would confound the two.

5.4. Main Results

Table 3 reports main-viewport performance (IR and Intersection@ δ) on the standard test set.

Table 4 reports the multi-region metrics. These are the quantities the proposed formulation is designed to improve: the gap between $\text{I@}\delta$ and $\text{BoK@}\delta$ indicates how much of the human attention distribution is reachable by additional regions, and $\text{OC@}\delta$ indicates how many of the disagreeing observers the output actually serves. A single-region method has $|\hat{\mathcal{P}}_t| = 1$, so its $\text{BoK@}\delta$ equals its $\text{I@}\delta$ by definition.

5.5. Generalisation Tests

Following [? ?], we evaluate generalisation to unseen races, starting locations, and maps.

Table 4: Multi-region coverage, per-observer service, and temporal stability. Lower VD is better. **[TODO: Fill in experimental values.]**

Method	BoK@0.5 $\uparrow$	OC@0.5 $\uparrow$	$ \hat{\mathcal{P}}_t $	VD (tiles) $\downarrow$
Mask R-CNN + prior (single)	[TODO: X]	[TODO: X]	1	[TODO: X]
CenterNet (single)	[TODO: X]	[TODO: X]	1	[TODO: X]
Director-CenterNet	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]

Table 5: Generalisation test: IR (4/4 game progression) for unseen conditions. **[TODO: Fill in experimental values.]**

Method	Unseen Race	Unseen Location	Unseen Map
Mask R-CNN + ROCI	0.615	0.559	0.581
Mask R-CNN + prior	[TODO: X]	[TODO: X]	[TODO: X]
Director-CenterNet	[TODO: X]	[TODO: X]	[TODO: X]

5.6. Ablation Study

Design of the lattice.. Four objectives are designed for this task, but only three of them are ablation axes. $\mathcal{L}_{\text{off}}$ and $\mathcal{L}_{\text{size}}$ are the standard CenterNet regression losses and are a precondition for decoding a box at all, not a contribution whose effect is measured. $\mathcal{L}_{\text{hcm}}$ is the base rather than an axis: setting $\lambda_{\text{hcm}} = 0$ leaves no positive supervision anywhere on the heatmap, because $\mathcal{L}_{\text{rmc}}$'s domain is restricted to Ψ_t and supervises neither the primary cell nor the background. The lattice is therefore $2^3 = 8$ rather than $2^4 = 16$.

Two of the eight cells are not measured, and for different reasons. $\mathcal{L}_{\text{hcm}} + \mathcal{L}_{\text{rep}}$ is undefined by construction: as noted in Section 4.3, $\mathcal{L}_{\text{rep}}$ penalises overlap between regions that already exist and cannot create any, so without $\mathcal{L}_{\text{rmc}}$ there is effectively one region per frame and nothing to repel. The effect of $\mathcal{L}_{\text{rep}}$ is thus defined only conditional on $\mathcal{L}_{\text{rmc}}$, and the empty cell records that dependency rather than an omission. $\mathcal{L}_{\text{hcm}} + \mathcal{L}_{\text{smooth}}$ is omitted as redundant: the effect of the smoothness objective is read directly from VD between the third and fourth rows of Table 6.

Table 6 reports the cumulative series and Table 7 the leave-one-out series; the third row of the former and the last row of the latter are the same configuration.

Table 6: Cumulative ablation (fold 1, 4/4 game progression, mean over seeds {123, 456, 789}). ✓ = included. **[TODO: Fill in experimental values.]**

$\mathcal{L}_{\text{hcm}}$	$\mathcal{L}_{\text{rmc}}$	$\mathcal{L}_{\text{rep}}$	$\mathcal{L}_{\text{smooth}}$	IR ↑	OC@0.5 ↑	$ \hat{\mathcal{P}}_t $	VD ↓
✓				[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
✓	✓			[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
✓	✓	✓		[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
✓	✓	✓	✓	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]

Table 7: Leave-one-out ablation (fold 1, 4/4 game progression, mean over seeds {123, 456, 789}). The $\mathcal{L}_{\text{hcm}} + \mathcal{L}_{\text{rep}}$ cell of the lattice is omitted because $\mathcal{L}_{\text{rep}}$ has no effect without $\mathcal{L}_{\text{rmc}}$. **[TODO: Fill in experimental values.]**

Configuration	IR ↑	OC@0.5 ↑	$ \hat{\mathcal{P}}_t $	VD ↓
Full	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
w/o $\mathcal{L}_{\text{rmc}}$	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
w/o $\mathcal{L}_{\text{rep}}$	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]
w/o $\mathcal{L}_{\text{smooth}}$	[TODO: X]	[TODO: X]	[TODO: X]	[TODO: X]

Reporting. $|\hat{\mathcal{P}}_t|$ denotes the mean number of regions emitted per frame, which we report alongside the accuracy metrics because $\mathcal{L}_{\text{rmc}}$ and $\mathcal{L}_{\text{rep}}$ are not interpretable when it is close to one: with a single region per frame, OC@ δ degenerates to the single-region case and $\mathcal{L}_{\text{rep}}$ has no argument. We also report the seed standard deviation, since several of the differences between neighbouring rows are of the same order as run-to-run variation. **[TODO: Report seed std once the three seeds complete.]**

5.7. Qualitative Analysis

[TODO: Insert qualitative figures: (1) Side-by-side: single-viewport vs. Director-CenterNet main + PIP during concurrent game events (battle + resource harassment). (2) Heatmap visualisation: aggregated observer coverage a_t , its smoothed field with the extracted ranked modes marked, the predicted heatmap $\hat{\mathcal{Y}}_t$, and the extracted region centres. (3) Temporal trajectory plots: frame-by-frame main viewport centre position for Mask R-CNN + prior vs. Director-CenterNet, illustrating reduced VD.]

6. Discussion

6.1. *Why Multi-Region Finding Matters for Esports Broadcasting*

The fundamental tension in Esports broadcasting is that a single camera cannot cover the full extent of a large, concurrent game state. Real-time strategy games such as StarCraft exhibit up to 40 times the unit count of MOBA games [?], making multi-region coverage not merely a convenience but a broadcast necessity. Director-CenterNet addresses this need directly, and does so using supervision that was already present in the data prior work collected: the minority modes of the observer distribution that ROCI and fixed-kernel priors were designed to suppress. No new form of annotation is introduced, only a denser sampling of the same protocol.

6.2. *CenterNet vs. Mask R-CNN for MRVP*

CenterNet offers three structural advantages over Mask R-CNN in the MRVP setting. First, its dense heatmap retains the multimodal structure of the target, whereas an instance mask forces a discrete assignment that discards it. Second, local-maximum extraction is more stable and efficient than repeated non-maximum suppression over proposal boxes, and it yields a variable number of regions without changing the model. Third, the support-weighted Gaussian rendering provides soft supervision that encodes the *ranking* of human attention, not merely its location.

6.3. *Heatmap Resolution*

The finest FPN level gives a stride of 4, so a 12×20 -tile viewport spans only 3×5 heatmap cells. Relative to target size this is considerably coarser than the regime CenterNet was designed for, where objects span tens of cells. Positional precision is not the binding constraint, since the offset head decodes sub-cell position continuously; what a coarse stride limits is the density of candidate centres and the separability of nearby modes, both of which bear on the multi-region metrics rather than on IR. We leave a systematic study of heatmap resolution to future work, and note that several constants in the method — the non-maximum suppression kernel in particular, whose radius must remain commensurate with the minimum mode separation D — are specified in tile units and would have to be rescaled alongside the stride.

6.4. Toward Domain-Agnostic Multi-Region Observers

The MRVP formulation and Director-CenterNet architecture are designed to be domain-agnostic. The structured game-state tensor can be replaced by any multi-channel spatial representation (e.g., MOBA minimap channels, drone swarm positional heatmaps, multi-CCTV occupancy grids). The four-loss framework remains applicable wherever observational data from several human operators is available, since the ranked modes that supervise the auxiliary regions are derived from disagreement among those operators rather than from any game-specific quantity. Where only a single operator’s data exists, the modes collapse and the formulation degenerates to the single-region case.

6.5. Limitations

Track B (zero human-GT setting). Director-CenterNet currently operates in Track A (human observation data available for the main viewport). Extending to Track B — where no human viewport data exists and the system must rely entirely on unsupervised saliency signals — is a critical direction for cross-game generalisation and is left as future work.

Temporal modelling. $\mathcal{L}_{\text{smooth}}$ acts on pairs of consecutive frames. Incorporating longer-range temporal context via recurrent modules or temporal self-attention may further reduce thrashing for events that evolve over several seconds.

PIP count. We fix $K = 3$ in all experiments. Adapting K dynamically beyond the confidence-threshold mechanism could improve coverage when more than two concurrent events occur simultaneously.

Live integration. All experiments are conducted in offline replay mode. Real-time integration requires addressing the pipeline latency between game-state extraction, inference, and camera actuation.

7. Conclusion

We presented Director-CenterNet, a multi-region viewport prediction framework for automated Esports broadcasting. By reformulating automatic observing as Multi-Region Viewport Prediction and introducing four complementary training losses — Human Consensus Match, Ranked Mode Coverage, Spatial Repulsion, and Trajectory Smoothness — Director-CenterNet

predicts an ordered set of presentation-agnostic regions from multi-channel game-state tensors in a single forward pass. The Ranked Mode Coverage loss supervises the auxiliary regions with the minority observer attention that previous work discarded as noise, so no new form of annotation is required, while the Spatial Repulsion and Trajectory Smoothness losses remove redundant overlap and camera thrashing respectively. Experiments on StarCraft: Brood War benchmarks demonstrate improvements in primary-region overlap, multi-region coverage, per-observer service, and temporal stability compared to single-region baselines.

Future work will explore Track B (zero human-GT) operation via unsupervised saliency estimation, longer-range temporal context modelling, and extension to MOBA games and non-game surveillance domains.

CRedit authorship contribution statement

[**TODO:** Fill in author contributions following CRedit taxonomy.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

[**TODO:** Add funding acknowledgements.]

Appendix A. Mode Extraction Procedure

Algorithm-level detail for Section 3.3. Given the U observer viewports at frame t , we form the coverage map a_t of Eq. 4, smooth it with a Gaussian filter of width σ_m under zero padding, and retain grid points that are local maxima of the smoothed field under a 3×3 maximum filter and that exceed $\theta \cdot \max \tilde{a}_t$. Candidates are visited in descending order of $\tilde{a}_t$ and a candidate is kept only if it lies at least D tiles from every mode already kept. Each kept mode receives the support of Eq. 5, and modes are ordered by support with ties broken by smoothed response. At most M modes are retained per frame.

The minimum separation D is the parameter that decides what counts as two distinct modes rather than one broad region of agreement, and it interacts with the extraction stride: the non-maximum suppression applied to the predicted heatmap must have a radius no larger than $D/2$ in tile units, or a single mode yields several peaks within its own viewport and the region count is inflated by duplicates.

Appendix B. Hyperparameter Summary

References

Table B.8: Hyperparameter settings used in all experiments unless otherwise stated.

Hyperparameter	Value	Description
Backbone	ResNet-50	Pretrained on ImageNet
FPN stride s	4	Finest FPN level; 32×32 heatmap
Map size	128×128 tiles	Consumed at native resolution
Viewport (h, w)	12×20 tiles	3×5 heatmap cells
Gaussian σ	2 tiles	GT rendering, on the assigned cell
Stacking Δ	8 frames	Temporal stride
Stacking W	3	Number of past frames
Mode σ_m	4 tiles	Coverage-map smoothing
Mode θ	0.35	Relative peak floor
Mode D	12 tiles	Minimum mode separation
Mode M	5	Max modes retained per frame
λ_{off}	1.0	Offset loss weight
λ_{sz}	0.1	Size loss weight
λ_{rmc}	0.5	Ranked mode coverage weight
λ_{rep}	0.3	Repulsion loss weight
λ_{sm}	0.02	Smoothness loss weight
Δ_s	5 tiles	Huber knee for $\mathcal{L}_{\text{smooth}}$
E_0, E_1	5, 10	Smoothness warmup start / full
r	2 cells	Soft-centre window radius
ϵ	0.01	Auxiliary support floor
K	3	Size of the predicted region set
τ	0.1	Peak confidence threshold ($< 1/U$)
Optimiser	SGD	momentum 0.9
Learning rate	5×10^{-3}	StepLR (step 3, γ 0.1)
Weight decay	5×10^{-4}	
Batch size	16	
Epochs	30	